# DPEN022: Enabling Mathematics B
# Solutions to Class Questions Week 2

## Week 2A

### Trigonometry - Trigonometric Identities

1. $$(\sin 60)^2 + (\cos 60)^2 = \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2$$
$$= \frac{3}{4} + \frac{1}{4} = 1$$

2. (i) $\sin^2\theta + \cos^2\theta = 1$

$$\cos^2\theta = 1 - \sin^2\theta$$
$$= 1 - \left(\frac{2}{\sqrt{5}}\right)^2$$
$$= \frac{1}{5}$$
$$\cos\theta = \pm\frac{1}{\sqrt{5}}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$
$$= \frac{2}{\sqrt{5}} \div \frac{1}{\sqrt{5}}$$
$$= 2$$

Since an acute angle is in Q1, cosine is positive. $\therefore \cos\theta = \frac{1}{\sqrt{5}}$

(ii) $\sin^2\theta + \cos^2\theta = 1$

$$\sin^2\beta = 1 - \cos^2\beta$$
$$= 1 - \left(\frac{-2}{3}\right)^2$$
$$= \frac{5}{9}$$

$$\tan\beta = \frac{\sin\beta}{\cos\beta}$$
$$= \frac{\sqrt{5}}{3} \div -\frac{2}{3} = -\frac{\sqrt{5}}{2}$$

$\sin\beta = \pm\frac{\sqrt{5}}{3}$ Since an obtuse angle is in Q2, sine is positive. $\therefore \sin\beta = \frac{\sqrt{5}}{3}$

(iii) $\sec^2\theta = 1 + \tan^2\theta$

$$\sec^2 A = 1 + \tan^2 A$$
$$= 1 + \left(\frac{1}{3}\right)^2$$
$$= \frac{9+1}{9} = \frac{10}{9}$$

$\sec A = \pm\frac{\sqrt{10}}{3}$ Since reflex angle $A$ has a positive tangent, it must be in Q3.

Cosine and secant are negative in Q3. $\therefore \sec A = -\frac{\sqrt{10}}{3}$ and $\cos A = -\frac{3}{\sqrt{10}}$

3. Since sine is negative, the angle must be in Q3 or Q4.
Cosine is negative in Q3 and positive in Q4.

$$\cos^2\theta = 1-\sin^2\theta$$
$$= 1-\left(-\frac{1}{5}\right)^2$$
$$= 1-\frac{1}{25} = \frac{24}{25}$$

$$\cos\theta = \pm\frac{\sqrt{24}}{\sqrt{5}}$$
$$= \pm\frac{2\sqrt{6}}{\sqrt{5}}$$

4. (i) $2-2\cos^2\theta = 2(1-\cos^2\theta)$
$= 2\sin^2\theta$

(ii) $\sec^2\theta(1-\cos^2\theta) = \frac{1}{\cos^2\theta}\times\sin^2\theta$
$= \frac{\sin^2\theta}{\cos^2\theta}$
$= \tan^2\theta$

(iii) $\tan\theta\cos\theta = \frac{\sin\theta}{\cos\theta}\cos\theta = \sin\theta$

(iv) $1-\sin^2\theta-\cos^2\theta-1 = -(\sin^2\theta+\cos^2\theta)$
$= -1$

(v) $\frac{\tan\theta\sec\theta}{\sec^2\theta} = \frac{\tan\theta}{\sec\theta}$
$= \tan\theta\div\frac{1}{\cos\theta}$
$= \frac{\sin\theta}{\cos\theta}\times\frac{\cos\theta}{1}$
$= \sin\theta$

(vi) $\tan\theta\sqrt{1-\sin^2\theta} = \tan\theta\sqrt{\cos^2\theta}$
$= \tan\theta\cos\theta$
$= \frac{\sin\theta}{\cos\theta}\cos\theta$
$= \sin\theta$

(vii)
$$\frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta} = \frac{\sin\theta}{\cos\theta}\frac{\sin\theta}{\sin\theta}+\frac{\cos\theta}{\sin\theta}\frac{\cos\theta}{\cos\theta}$$
$$= \frac{\sin^2\theta}{\sin\theta\cos\theta}+\frac{\cos^2\theta}{\sin\theta\cos\theta}$$
$$= \frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}$$
$$= \frac{1}{\sin\theta\cos\theta}$$
$$= \frac{1}{\sin\theta}\frac{1}{\cos\theta}$$
$$= \cos ec\theta\sec\theta$$

5. (i)
$$\frac{2\cos^2\theta}{1-\sin\theta} = 2+2\sin\theta$$
$$LHS = \frac{2\cos^2\theta}{1-\sin\theta}$$
$$= \frac{2\cos^2\theta}{1-\sin\theta}\times\frac{1+\sin\theta}{1+\sin\theta}$$
$$= \frac{2\cos^2\theta(1+\sin\theta)}{1-\sin^2\theta}$$
$$= \frac{2\cos^2\theta(1+\sin\theta)}{\cos^2\theta}$$
$$= 2(1+\sin\theta)$$
$$= 2+2\sin\theta$$
$$= RHS$$

(ii)
$$(\sec^2\theta-1)\cos^2\theta = \sin^2\theta$$
$$LHS = (\sec^2\theta-1)\cos^2\theta$$
$$= \tan^2\theta\cos^2\theta$$
$$= \frac{\sin^2\theta}{\cos^2\theta}\times\cos^2\theta$$
$$= \sin^2\theta$$
$$= RHS$$

(iii)

$$\frac{\sin\theta}{1+\cos\theta}=\frac{1-\cos\theta}{\sin\theta}$$

$$\begin{aligned}
LHS &= \frac{\sin\theta}{1+\cos\theta}\\
&= \frac{\sin\theta}{1+\cos\theta}\times\frac{1-\cos\theta}{1-\cos\theta}\\
&= \frac{\sin\theta(1-\cos\theta)}{1-\cos^2\theta}\\
&= \frac{\sin\theta(1-\cos\theta)}{\sin^2\theta}\\
&= \frac{(1-\cos\theta)}{\sin\theta}\\
&= RHS
\end{aligned}$$

(iv)

$$\sin\theta\tan\theta=\frac{1-\cos^2\theta}{\cos\theta}$$

$$\begin{aligned}
LHS &= \sin\theta\tan\theta\\
&= \sin\theta\frac{\sin\theta}{\cos\theta}\\
&= \frac{\sin^2\theta}{\cos\theta}\\
&= \frac{1-\cos^2\theta}{\cos\theta}\\
&= RHS
\end{aligned}$$

6. (i)

$$\begin{aligned}
2\sin^2 x &= \cos x+2\\
2(1-\cos^2 x) &= \cos x+2\\
2-2\cos^2 x &= \cos x+2\\
-2\cos^2 x &= \cos x\\
2\cos^2 x+\cos x &= 0\\
\cos x(2\cos x+1) &= 0
\end{aligned}$$

$$\cos x=0 \qquad \cos x=-0.5$$

$$x=90^\circ,270^\circ \qquad x=180^\circ\pm 60$$

$$=120^\circ,240^\circ$$

$$x=90^\circ,120^\circ,240^\circ,270^\circ$$

(ii)

$$\begin{aligned}
3-3\sin x &= \cos^2 x\\
3-3\sin x &= (1-\sin^2 x)\\
\sin^2 x-3\sin x+2 &= 0\\
(\sin x-2)(\sin x-1) &= 0
\end{aligned}$$

$$\sin x\neq 2$$

$$\sin x=1$$

$$x=90^\circ$$

(iii)

$$\begin{aligned}
\tan^2 x+2\sec^2 x-3 &= 0\\
\tan^2 x+2(1+\tan^2 x)-3 &= 0\\
\tan^2 x+2\tan^2 x+2-3 &= 0\\
3\tan^2 x-1 &= 0\\
\tan^2 x &= \frac{1}{3}
\end{aligned}$$

$$\tan x=\pm\frac{1}{\sqrt{3}}$$

$$x=30,180\pm30,360-30$$

$$=30^\circ,150^\circ,210^\circ,330^\circ$$

**<u>Week 2D</u>**

**Limits**

1. (i) 2 (ii) DNE
   (iii) 1 (iv) 0

2. (i) $-2$ & 3
   (ii) VA $x=-2$
   (iii) HA $y=1$

3. (i)
$$\lim_{x\to 2} 5x-3 = 5(2)-3 = 7$$

(ii)
$$\lim_{x\to 1}\frac{x^2-9}{x^2+1}=\frac{1^2-9}{1^2+1} = -4$$

(iii) $\lim_{x\to 3}\frac{3-x}{x}=\frac{3-3}{3}=0$

(iv) $\lim_{x\to 1}3^{2x-1}=3^{2(1)-1}=3^1=3$

(v) $\lim_{x\to 0}e^x=e^0=1$

(vi) $\lim_{x\to 3}\frac{1}{2^{2x-1}}-5=\frac{1}{2^{2(3)-1}}-5=\frac{1}{32}-5=-\frac{159}{32}$

(vii)
$$\begin{aligned}\lim_{x\to 3}\left(\log_4(3x-5)\right) &= \log_4(3(3)-5)\\ &= \log_4(9-5)\\ &= \log_4(4)\\ &= 1\end{aligned}$$

(viii) $\lim_{x\to 2}\frac{\ln(x-1)}{(x-1)}=\frac{\ln(2-1)}{(2-1)}=\ln 1=0$

(ix) $\lim_{x\to \frac{\pi}{3}}\tan x=\tan(\tfrac{\pi}{3})=\sqrt{3}$

(x)
$$\begin{aligned}\lim_{x\to\frac{\pi}{6}}\cot 4x &= \frac{\cos 4(\frac{\pi}{6})}{\sin 4(\frac{\pi}{6})}=\frac{\cos\frac{2\pi}{3}}{\sin\frac{2\pi}{3}}\\ &= \frac{-\frac{1}{2}}{\frac{\sqrt{3}}{2}}=-\frac{1}{\sqrt{3}}\end{aligned}$$

(xi) This limit is in the form of $\frac{0}{0}$, therefore, factorise
$$\begin{aligned}\lim_{x\to 4}\frac{x^2-16}{x-4} &= \lim_{x\to 4}\frac{(x-4)(x+4)}{x-4}\\ &= \lim_{x\to 4}x+4\\ &= 4+4\\ &= 8\end{aligned}$$

(xii) This limit is in the form of $\frac{0}{0}$, therefore, factorise

Therefore factorise

$$\lim_{x\to 1}\frac{x^2-3x+2}{x-1}=\lim_{x\to 1}\frac{(x-1)(x-2)}{x-1}$$
$$=\lim_{x\to 1}x-2$$
$$=1-2$$
$$=-1$$

(xiii) This limit is in the form of $\frac{0}{0}$, therefore, factorise

$$\lim_{x\to 2}\frac{x^2+x-6}{x^2-4}=\lim_{x\to 2}\frac{(x+3)(x-2)}{(x-2)(x+2)}$$
$$=\lim_{x\to 2}\frac{(x+3)}{(x+2)}$$
$$=\frac{2+3}{2+2}=\frac{5}{4}$$

4.

| | | | | |
|---|---|---|---|---|
| Graph 1 | (i) | DNE | (ii) | DNE |
| Graph 2 | (i) | 3 | (ii) | 3 |
| Graph 3 | (i) | DNE | (ii) | DNE |

5. (i) This limit is in the form of $\frac{\infty}{\infty}$, therefore, divide each term by the highest power of $x$.

$$\lim_{x\to\infty}\frac{x^2+x-6}{2x^2-4}=\lim_{x\to\infty}\frac{\frac{x^2}{x^2}+\frac{x}{x^2}-\frac{6}{x^2}}{\frac{2x^2}{x^2}-\frac{4}{x^2}}$$
$$=\lim_{x\to\infty}\frac{1+\frac{1}{x}-\frac{6}{x^2}}{2-\frac{4}{x^2}}$$
$$=\frac{1+0-0}{2-0}$$
$$=\frac{1}{2}$$

(ii) This limit is in the form of $\frac{\infty}{\infty}$, therefore, divide each term by the highest power of $x$.

$$\lim_{x\to\infty}\frac{x-6}{x^2-4}=\lim_{x\to\infty}\frac{\frac{x}{x^2}-\frac{6}{x^2}}{\frac{x^2}{x^2}-\frac{4}{x^2}}$$

$$=\lim_{x\to\infty}\frac{\frac{1}{x}-\frac{6}{x^2}}{1-\frac{4}{x^2}}$$

$$=\frac{0-0}{1-0}$$

$$=\frac{0}{1}=0$$

(iii) This limit is in the form of $\frac{\infty}{\infty}$, therefore, divide each term by the highest power of $x$.

$$\lim_{x\to\infty}\frac{4x^3+x-8}{2x^5-4x+2}=\lim_{x\to\infty}\frac{\frac{4x^3}{x^5}+\frac{x}{x^5}-\frac{8}{x^5}}{\frac{2x^5}{x^5}-\frac{4x}{x^5}+\frac{2}{x^5}}$$

$$=\lim_{x\to\infty}\frac{\frac{4}{x^2}+\frac{1}{x^4}-\frac{8}{x^5}}{2-\frac{4}{x^4}+\frac{2}{x^5}}$$

$$=\frac{0+0-0}{2-0-0}$$

$$=\frac{0}{2}$$

$$=0$$

(iv) This limit is in the form of $\frac{\infty}{\infty}$, therefore, divide each term by the highest power of $x$.

$$\lim_{x\to\infty}\frac{7x^2+x-6}{2x^2-4}=\lim_{x\to\infty}\frac{\frac{7x^2}{x^2}+\frac{x}{x^2}-\frac{6}{x^2}}{\frac{2x^2}{x^2}-\frac{4}{x^2}}$$

$$=\lim_{x\to\infty}\frac{7+\frac{1}{x}-\frac{6}{x^2}}{2-\frac{4}{x^2}}$$

$$=\frac{7+0-0}{2-0}$$

$$=\frac{7}{2}$$